

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - IPDE alpha = 0	35.3	16.8	2.4	0.6	0.2	44.7
U2 - IPDE alpha = 0.3	32.7	12.1	1.6	0.3	0.1	53.2
U2 - IPDE alpha = 0.6	29.8	9.9	1.1	0	0	59.2
U2 - IPDE alpha = 0.9	26.4	7.1	0.3	0	0	66.2
U3 - IPDE alpha = 0	35.4	15.6	2	0	0.1	46.9
U3 - IPDE alpha = 0.3	32.4	11.4	1	0	0.1	55.1
U3 - IPDE alpha = 0.6	30.5	8.5	0.8	0	0	60.2
U3 - IPDE alpha = 0.9	26.9	5.2	0.5	0	0	67.4
Mean patients treated						
U2 - IPDE alpha = 0	14.6	6.3	2.5	1.4	0.4	
U2 - IPDE alpha = 0.3	15.1	6.1	1.9	1	0.2	
U2 - IPDE alpha = 0.6	15.5	5.7	1.4	0.6	0.1	
U2 - IPDE alpha = 0.9	15.9	5.1	1	0.4	0.1	
U3 - IPDE alpha = 0	15.1	5.9	2.3	1.4	0.4	
U3 - IPDE alpha = 0.3	15.5	5.7	1.9	0.9	0.2	
U3 - IPDE alpha = 0.6	15.8	5.4	1.4	0.6	0.1	
U3 - IPDE alpha = 0.9	16.2	4.8	1	0.4	0	
Mean IPDE patients						
U2 - IPDE alpha = 0	1	3.4	0.8	0.6	0.1	
U2 - IPDE alpha = 0.3	1	3.3	0.6	0.4	0.1	
U2 - IPDE alpha = 0.6	1	3.2	0.5	0.3	0	
U2 - IPDE alpha = 0.9	1	3	0.4	0.2	0	
U3 - IPDE alpha = 0	1	3.3	0.7	0.6	0.1	
U3 - IPDE alpha = 0.3	1	3.2	0.6	0.4	0.1	
U3 - IPDE alpha = 0.6	1	3.1	0.5	0.3	0	
U3 - IPDE alpha = 0.9	1	2.8	0.4	0.2	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - IPDE alpha = 0	25.2	18.3	39.3	13.9	2.3	1
U2 - IPDE alpha = 0.3	25.7	21.8	37.8	11.8	1.9	1
U2 - IPDE alpha = 0.6	29	24.8	36.2	8.5	0.8	0.7
U2 - IPDE alpha = 0.9	30.4	29.3	31.9	7.1	0.3	1
U3 - IPDE alpha = 0	30.8	19.1	35.4	11.9	1.8	1
U3 - IPDE alpha = 0.3	31.5	21	34.6	10.8	1.2	0.9
U3 - IPDE alpha = 0.6	31.7	25.5	34.5	7	0.5	0.8
U3 - IPDE alpha = 0.9	33	30	30.7	5.3	0.1	0.9
Mean patients treated						
U2 - IPDE alpha = 0	9.2	7	9	3.8	0.9	
U2 - IPDE alpha = 0.3	9.7	7.7	8.8	3.1	0.6	
U2 - IPDE alpha = 0.6	10.4	8.5	8.2	2.5	0.3	
U2 - IPDE alpha = 0.9	11	9.2	7.5	2	0.1	
U3 - IPDE alpha = 0	9.8	7.2	8.7	3.5	0.7	
U3 - IPDE alpha = 0.3	10.3	7.7	8.5	2.9	0.5	
U3 - IPDE alpha = 0.6	10.9	8.7	7.7	2.3	0.2	
U3 - IPDE alpha = 0.9	11.5	9.4	7.1	1.8	0.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.5	3.2	1.7	1	0.2	
U2 - IPDE alpha = 0.3	0.6	3.4	1.7	0.8	0.2	
U2 - IPDE alpha = 0.6	0.7	3.7	1.7	0.7	0.1	
U2 - IPDE alpha = 0.9	0.7	3.9	1.8	0.6	0.1	
U3 - IPDE alpha = 0	0.6	3.2	1.6	0.8	0.2	
U3 - IPDE alpha = 0.3	0.6	3.4	1.6	0.8	0.1	
U3 - IPDE alpha = 0.6	0.7	3.7	1.6	0.6	0.1	
U3 - IPDE alpha = 0.9	0.8	3.9	1.6	0.5	0.1	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - IPDE alpha = 0	9.6	46.7	26.9	12.3	2.1	2.4
U2 - IPDE alpha = 0.3	10.3	49.7	25.7	10	1.6	2.7
U2 - IPDE alpha = 0.6	11.8	51.6	25.3	7.5	0.8	3
U2 - IPDE alpha = 0.9	13	54.7	23.4	5.2	0.1	3.6
U3 - IPDE alpha = 0	12.4	53.6	21.1	8.7	1.7	2.5
U3 - IPDE alpha = 0.3	12.5	54.8	20.9	7.9	1.2	2.7
U3 - IPDE alpha = 0.6	13.1	57.2	21.2	5	0.6	2.9
U3 - IPDE alpha = 0.9	14.1	57.9	20	4.4	0	3.6
Mean patients treated						
U2 - IPDE alpha = 0	6.6	11.3	6.8	3.8	1.1	
U2 - IPDE alpha = 0.3	7.2	12	6.7	3.1	0.8	
U2 - IPDE alpha = 0.6	8	12.4	6.4	2.5	0.4	
U2 - IPDE alpha = 0.9	8.8	12.7	6	1.9	0.2	
U3 - IPDE alpha = 0	6.8	12.2	6.4	3.4	0.9	
U3 - IPDE alpha = 0.3	7.4	12.7	6.1	2.9	0.6	
U3 - IPDE alpha = 0.6	8.2	13.2	5.8	2.2	0.3	
U3 - IPDE alpha = 0.9	9.1	13.3	5.4	1.7	0.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.4	3.8	1.8	1.1	0.3	
U2 - IPDE alpha = 0.3	0.4	4	1.8	1	0.3	
U2 - IPDE alpha = 0.6	0.4	4.3	1.8	0.8	0.2	
U2 - IPDE alpha = 0.9	0.5	4.5	1.7	0.6	0.1	
U3 - IPDE alpha = 0	0.4	3.8	1.6	1	0.2	
U3 - IPDE alpha = 0.3	0.4	4	1.6	0.9	0.2	
U3 - IPDE alpha = 0.6	0.5	4.3	1.6	0.7	0.1	
U3 - IPDE alpha = 0.9	0.5	4.5	1.6	0.5	0.1	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - IPDE alpha = 0	6.7	6.6	17.5	52	17	0.2
U2 - IPDE alpha = 0.3	6.2	6.9	19.1	53.9	13.6	0.3
U2 - IPDE alpha = 0.6	6.4	8.4	21.1	53.3	10.6	0.2
U2 - IPDE alpha = 0.9	8.3	8.7	25	49.8	7.9	0.3
U3 - IPDE alpha = 0	9.6	7.8	16.9	52.1	13.4	0.2
U3 - IPDE alpha = 0.3	9.7	7.8	19.7	51.3	11.3	0.2
U3 - IPDE alpha = 0.6	9.7	9.9	20.3	51.6	8.3	0.2
U3 - IPDE alpha = 0.9	10.9	11.7	21.8	48.6	6.8	0.2
Mean patients treated						
U2 - IPDE alpha = 0	4.2	4.6	6.7	10.6	3.9	
U2 - IPDE alpha = 0.3	4.3	4.8	7.1	10.6	3.2	
U2 - IPDE alpha = 0.6	4.5	5.1	7.4	10.4	2.6	
U2 - IPDE alpha = 0.9	4.6	5.5	7.8	10	2.1	
U3 - IPDE alpha = 0	4.5	4.6	6.8	10.7	3.4	
U3 - IPDE alpha = 0.3	4.6	4.9	7.1	10.5	2.8	
U3 - IPDE alpha = 0.6	4.8	5.1	7.4	10.4	2.3	
U3 - IPDE alpha = 0.9	5	5.5	7.7	10	1.8	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.2	3	1.5	2.4	0.8	
U2 - IPDE alpha = 0.3	0.2	3	1.6	2.5	0.7	
U2 - IPDE alpha = 0.6	0.3	3.1	1.7	2.5	0.7	
U2 - IPDE alpha = 0.9	0.3	3.2	1.8	2.5	0.6	
U3 - IPDE alpha = 0	0.2	3	1.5	2.4	0.7	
U3 - IPDE alpha = 0.3	0.3	3	1.6	2.4	0.6	
U3 - IPDE alpha = 0.6	0.3	3.1	1.7	2.4	0.5	
U3 - IPDE alpha = 0.9	0.3	3.2	1.8	2.5	0.5	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - IPDE alpha = 0	5.5	38.4	25.4	19.7	10.7	0.3
U2 - IPDE alpha = 0.3	5	35.8	24.9	24.2	9.7	0.4
U2 - IPDE alpha = 0.6	4.5	35.9	26.5	23.3	9.4	0.4
U2 - IPDE alpha = 0.9	4.4	34.6	27.5	24.5	8.5	0.5
U3 - IPDE alpha = 0	8.1	43.6	24	17.6	6.4	0.3
U3 - IPDE alpha = 0.3	7.4	44.6	22.5	19.1	6	0.4
U3 - IPDE alpha = 0.6	7.4	43.1	25.2	17.9	6	0.4
U3 - IPDE alpha = 0.9	6.6	43.3	25.8	18.5	5.3	0.5
Mean patients treated						
U2 - IPDE alpha = 0	3.9	7.3	7	7.1	4.7	
U2 - IPDE alpha = 0.3	4	7.6	7.2	7.2	4	
U2 - IPDE alpha = 0.6	4.1	8.1	7.3	7	3.5	
U2 - IPDE alpha = 0.9	4.3	8.4	7.5	6.9	2.9	
U3 - IPDE alpha = 0	4	7.9	7.3	6.9	3.9	
U3 - IPDE alpha = 0.3	4.2	8.5	7.3	6.8	3.3	
U3 - IPDE alpha = 0.6	4.3	8.8	7.4	6.5	2.8	
U3 - IPDE alpha = 0.9	4.5	9.1	7.6	6.5	2.3	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.1	3.1	1.9	2.2	1.2	
U2 - IPDE alpha = 0.3	0.1	3.3	2	2.3	1.1	
U2 - IPDE alpha = 0.6	0.2	3.3	2	2.1	1.1	
U2 - IPDE alpha = 0.9	0.2	3.5	2.1	2.1	1	
U3 - IPDE alpha = 0	0.2	3.2	2	2.2	0.9	
U3 - IPDE alpha = 0.3	0.2	3.3	2	2.1	0.9	
U3 - IPDE alpha = 0.6	0.2	3.4	2.1	2	0.9	
U3 - IPDE alpha = 0.9	0.2	3.5	2.1	2	0.8	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - IPDE alpha = 0	63.5	17.5	4.9	1.9	0.5	11.7
U2 - IPDE alpha = 0.3	65.9	15.8	3.6	1	0.2	13.5
U2 - IPDE alpha = 0.6	65.4	16.4	1.9	1	0.1	15.2
U2 - IPDE alpha = 0.9	66.2	14.5	1.5	0.3	0	17.5
U3 - IPDE alpha = 0	66.7	16.2	3.7	1.2	0.5	11.7
U3 - IPDE alpha = 0.3	66.8	15.3	3.6	0.7	0.2	13.4
U3 - IPDE alpha = 0.6	67.1	15.7	1.4	0.7	0.1	15
U3 - IPDE alpha = 0.9	69.4	12.4	0.5	0.3	0	17.4
Mean patients treated						
U2 - IPDE alpha = 0	17.5	6.1	2.7	1.3	0.3	
U2 - IPDE alpha = 0.3	18.4	6	2.2	0.8	0.2	
U2 - IPDE alpha = 0.6	19	6.2	1.6	0.6	0.1	
U2 - IPDE alpha = 0.9	19.8	5.9	1.1	0.2	0	
U3 - IPDE alpha = 0	18.1	6	2.4	1.2	0.3	
U3 - IPDE alpha = 0.3	18.8	6	1.9	0.8	0.2	
U3 - IPDE alpha = 0.6	19.4	6.1	1.3	0.5	0.1	
U3 - IPDE alpha = 0.9	20.1	5.8	0.9	0.2	0	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.7	2.6	0.7	0.4	0.1	
U2 - IPDE alpha = 0.3	0.7	2.6	0.6	0.3	0	
U2 - IPDE alpha = 0.6	0.7	2.7	0.5	0.2	0	
U2 - IPDE alpha = 0.9	0.7	2.7	0.4	0.1	0	
U3 - IPDE alpha = 0	0.7	2.5	0.6	0.3	0.1	
U3 - IPDE alpha = 0.3	0.8	2.5	0.5	0.2	0	
U3 - IPDE alpha = 0.6	0.8	2.6	0.4	0.2	0	
U3 - IPDE alpha = 0.9	0.8	2.6	0.3	0.1	0	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - IPDE alpha = 0	13.1	48.1	22.8	11.5	3.5	1
U2 - IPDE alpha = 0.3	12.3	50.7	24.8	8.7	2.3	1.2
U2 - IPDE alpha = 0.6	13.2	55.8	22.4	6.8	0.5	1.3
U2 - IPDE alpha = 0.9	15.1	59.8	20	3.3	0.6	1.2
U3 - IPDE alpha = 0	15.5	52.7	20	9.1	1.7	1
U3 - IPDE alpha = 0.3	14.4	56.3	20.7	6.5	1.1	1
U3 - IPDE alpha = 0.6	15.3	59.6	19.6	4.1	0.3	1.1
U3 - IPDE alpha = 0.9	16.4	62.5	16.9	2.6	0.4	1.2
Mean patients treated						
U2 - IPDE alpha = 0	7.6	12.6	5.8	2.9	0.9	
U2 - IPDE alpha = 0.3	8	13.4	5.8	2.2	0.5	
U2 - IPDE alpha = 0.6	8.6	14.5	5.1	1.4	0.2	
U2 - IPDE alpha = 0.9	9.5	15	4.4	0.8	0.1	
U3 - IPDE alpha = 0	8	13.5	5.3	2.5	0.6	
U3 - IPDE alpha = 0.3	8.3	14.4	5	1.8	0.3	
U3 - IPDE alpha = 0.6	9	15.1	4.5	1.1	0.1	
U3 - IPDE alpha = 0.9	9.8	15.5	3.8	0.7	0.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.3	3.4	1.3	0.7	0.2	
U2 - IPDE alpha = 0.3	0.3	3.6	1.3	0.6	0.1	
U2 - IPDE alpha = 0.6	0.4	3.9	1.3	0.4	0.1	
U2 - IPDE alpha = 0.9	0.4	4.1	1.1	0.3	0	
U3 - IPDE alpha = 0	0.3	3.5	1.1	0.6	0.1	
U3 - IPDE alpha = 0.3	0.4	3.6	1.1	0.4	0.1	
U3 - IPDE alpha = 0.6	0.5	3.9	1	0.3	0	
U3 - IPDE alpha = 0.9	0.5	4.2	0.9	0.2	0	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - IPDE alpha = 0	44.6	16.2	3.1	0.9	0.3	34.9
U2 - IPDE alpha = 0.3	44.8	12.3	1.6	0.7	0.3	40.3
U2 - IPDE alpha = 0.6	44.6	9.5	0.9	0.1	0.1	44.8
U2 - IPDE alpha = 0.9	44.4	5.7	0.6	0.1	0.1	49.1
U3 - IPDE alpha = 0	45.9	15.4	2.3	0.9	0.3	35.2
U3 - IPDE alpha = 0.3	45.1	11.7	1.7	0.4	0.2	40.9
U3 - IPDE alpha = 0.6	46.6	8	0.9	0.1	0	44.4
U3 - IPDE alpha = 0.9	44.9	5.5	0.5	0.1	0	49
Mean patients treated						
U2 - IPDE alpha = 0	15.7	6.2	2.3	1.2	0.3	
U2 - IPDE alpha = 0.3	16.4	5.8	1.8	0.7	0.2	
U2 - IPDE alpha = 0.6	17.1	5.4	1.2	0.4	0.1	
U2 - IPDE alpha = 0.9	17.5	4.9	0.8	0.2	0	
U3 - IPDE alpha = 0	16	6	2.2	1.1	0.3	
U3 - IPDE alpha = 0.3	16.7	5.6	1.7	0.7	0.2	
U3 - IPDE alpha = 0.6	17.3	5.3	1.2	0.4	0.1	
U3 - IPDE alpha = 0.9	17.7	4.7	0.8	0.2	0	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.9	2.9	0.7	0.4	0.1	
U2 - IPDE alpha = 0.3	0.9	2.9	0.5	0.3	0	
U2 - IPDE alpha = 0.6	0.9	2.7	0.4	0.2	0	
U2 - IPDE alpha = 0.9	0.9	2.5	0.3	0.1	0	
U3 - IPDE alpha = 0	0.9	2.9	0.6	0.4	0.1	
U3 - IPDE alpha = 0.3	0.9	2.8	0.5	0.3	0	
U3 - IPDE alpha = 0.6	0.9	2.6	0.4	0.2	0	
U3 - IPDE alpha = 0.9	0.8	2.5	0.3	0.1	0	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - IPDE alpha = 0	14.7	15.3	22.3	16.8	26.5	4.4
U2 - IPDE alpha = 0.3	15.9	16.6	23	17.5	21.7	5.3
U2 - IPDE alpha = 0.6	18.9	20.1	23.3	14.5	16.2	7
U2 - IPDE alpha = 0.9	20	23	22.6	13.2	11.6	9.6
U3 - IPDE alpha = 0	21.7	16.8	18.6	15.2	22.9	4.8
U3 - IPDE alpha = 0.3	22	20.7	18.6	15	18.3	5.4
U3 - IPDE alpha = 0.6	25.9	22.3	18.6	12.8	13.6	6.8
U3 - IPDE alpha = 0.9	24.6	24.6	19.9	11.9	9.6	9.4
Mean patients treated						
U2 - IPDE alpha = 0	7	6.1	5.9	5.5	5.1	
U2 - IPDE alpha = 0.3	7.6	6.4	6.1	5.1	4.2	
U2 - IPDE alpha = 0.6	8.3	7	5.9	4.9	3.2	
U2 - IPDE alpha = 0.9	9.2	7.5	5.8	4.3	2.5	
U3 - IPDE alpha = 0	7.6	6	5.8	5.4	4.8	
U3 - IPDE alpha = 0.3	8.2	6.3	5.9	5.2	3.9	
U3 - IPDE alpha = 0.6	9	6.8	5.8	4.8	3	
U3 - IPDE alpha = 0.9	9.8	7.3	5.6	4.3	2.2	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.5	3.6	1.6	1.9	1	
U2 - IPDE alpha = 0.3	0.6	3.8	1.6	1.8	0.9	
U2 - IPDE alpha = 0.6	0.6	4	1.7	1.6	0.7	
U2 - IPDE alpha = 0.9	0.7	4.2	1.7	1.5	0.6	
U3 - IPDE alpha = 0	0.6	3.5	1.5	1.8	0.9	
U3 - IPDE alpha = 0.3	0.6	3.8	1.6	1.7	0.8	
U3 - IPDE alpha = 0.6	0.7	3.9	1.6	1.6	0.7	
U3 - IPDE alpha = 0.9	0.8	4.1	1.6	1.5	0.6	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - IPDE alpha = 0	10.8	43.5	17.2	6.3	1.5	20.7
U2 - IPDE alpha = 0.3	14	41.2	15.1	4.2	0.9	24.6
U2 - IPDE alpha = 0.6	18.4	38.3	12.4	2.9	0.4	27.6
U2 - IPDE alpha = 0.9	21.3	34	9.3	1.4	0.1	33.9
U3 - IPDE alpha = 0	12.1	46	15.5	4.7	0.9	20.8
U3 - IPDE alpha = 0.3	15.9	41.4	13	3.7	0.8	25.2
U3 - IPDE alpha = 0.6	19.4	38.3	11.3	2.3	0.3	28.4
U3 - IPDE alpha = 0.9	22.2	33.4	9	1	0.1	34.3
Mean patients treated						
U2 - IPDE alpha = 0	9.5	10.7	4.8	2.2	0.7	
U2 - IPDE alpha = 0.3	10.7	10.5	4.3	1.7	0.4	
U2 - IPDE alpha = 0.6	11.9	10.2	3.8	1.2	0.2	
U2 - IPDE alpha = 0.9	13	9.8	3.1	0.7	0.1	
U3 - IPDE alpha = 0	9.7	11.1	4.6	2	0.5	
U3 - IPDE alpha = 0.3	10.9	10.9	3.9	1.5	0.3	
U3 - IPDE alpha = 0.6	11.9	10.4	3.5	1.1	0.1	
U3 - IPDE alpha = 0.9	13.3	10	2.8	0.7	0.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.5	4	1.1	0.7	0.2	
U2 - IPDE alpha = 0.3	0.5	4.2	1.1	0.5	0.1	
U2 - IPDE alpha = 0.6	0.6	4.3	1	0.4	0.1	
U2 - IPDE alpha = 0.9	0.6	4.4	0.9	0.3	0	
U3 - IPDE alpha = 0	0.5	4	1	0.6	0.1	
U3 - IPDE alpha = 0.3	0.6	4.2	1	0.5	0.1	
U3 - IPDE alpha = 0.6	0.6	4.2	0.9	0.4	0	
U3 - IPDE alpha = 0.9	0.6	4.3	0.8	0.2	0	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - IPDE alpha = 0	66	23.5	5.3	0.5	0	4.7
U2 - IPDE alpha = 0.3	66.3	23.3	4.2	0.1	0	6.1
U2 - IPDE alpha = 0.6	68.4	21.5	2.5	0.1	0	7.5
U2 - IPDE alpha = 0.9	67	22.4	1.8	0.1	0	8.7
U3 - IPDE alpha = 0	69.7	20.4	4	0.6	0	5.3
U3 - IPDE alpha = 0.3	68.2	22.6	2.7	0.2	0	6.3
U3 - IPDE alpha = 0.6	70.8	20	2	0.1	0	7.1
U3 - IPDE alpha = 0.9	70.2	20.5	1.6	0	0	7.7
Mean patients treated						
U2 - IPDE alpha = 0	16.5	7.9	3.4	1.4	0.2	
U2 - IPDE alpha = 0.3	17.5	7.7	2.9	1	0.1	
U2 - IPDE alpha = 0.6	18.1	7.7	2.4	0.8	0.1	
U2 - IPDE alpha = 0.9	18.7	7.8	1.8	0.5	0	
U3 - IPDE alpha = 0	17.1	7.6	3.2	1.3	0.2	
U3 - IPDE alpha = 0.3	18	7.6	2.5	0.9	0.1	
U3 - IPDE alpha = 0.6	18.7	7.5	2.1	0.7	0	
U3 - IPDE alpha = 0.9	19.2	7.6	1.7	0.5	0	
Mean IPDE patients						
U2 - IPDE alpha = 0	1	3.5	1	0.5	0.1	
U2 - IPDE alpha = 0.3	1	3.6	0.9	0.4	0	
U2 - IPDE alpha = 0.6	1.1	3.7	0.8	0.3	0	
U2 - IPDE alpha = 0.9	1	3.8	0.7	0.2	0	
U3 - IPDE alpha = 0	1.1	3.4	0.9	0.5	0.1	
U3 - IPDE alpha = 0.3	1	3.5	0.8	0.3	0	
U3 - IPDE alpha = 0.6	1.1	3.6	0.7	0.3	0	
U3 - IPDE alpha = 0.9	1.1	3.7	0.6	0.2	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - IPDE alpha = 0	11.7	11.2	8.4	44.2	24.1	0.4
U2 - IPDE alpha = 0.3	11.8	12.1	12.2	44.6	19	0.3
U2 - IPDE alpha = 0.6	11.7	16.3	11.4	44.7	15.6	0.3
U2 - IPDE alpha = 0.9	12.4	17.1	14.4	42.9	12.8	0.4
U3 - IPDE alpha = 0	16.6	14.3	6.7	41	21.2	0.2
U3 - IPDE alpha = 0.3	17.1	15.4	9.1	40.7	17.4	0.3
U3 - IPDE alpha = 0.6	16.9	16.2	10.5	41.7	14.4	0.3
U3 - IPDE alpha = 0.9	16.3	19.7	12.4	40.3	10.9	0.4
Mean patients treated						
U2 - IPDE alpha = 0	4.6	5.1	5.4	9.4	5.4	
U2 - IPDE alpha = 0.3	4.8	5.5	5.7	9.3	4.7	
U2 - IPDE alpha = 0.6	4.9	5.8	6.1	9.1	4.1	
U2 - IPDE alpha = 0.9	5.1	6.3	6.3	8.8	3.4	
U3 - IPDE alpha = 0	4.9	5.1	5.3	9.6	5.1	
U3 - IPDE alpha = 0.3	5.2	5.6	5.6	9.3	4.4	
U3 - IPDE alpha = 0.6	5.4	5.9	5.9	9	3.8	
U3 - IPDE alpha = 0.9	5.7	6.4	6.1	8.7	3.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.3	3.1	1.5	2.3	1.1	
U2 - IPDE alpha = 0.3	0.4	3.2	1.6	2.4	1.1	
U2 - IPDE alpha = 0.6	0.4	3.3	1.7	2.4	1	
U2 - IPDE alpha = 0.9	0.4	3.4	1.7	2.3	1	
U3 - IPDE alpha = 0	0.4	3.1	1.5	2.3	1	
U3 - IPDE alpha = 0.3	0.5	3.2	1.5	2.3	1	
U3 - IPDE alpha = 0.6	0.5	3.3	1.6	2.3	0.9	
U3 - IPDE alpha = 0.9	0.5	3.5	1.7	2.2	0.8	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - IPDE alpha = 0	12.2	44.8	17.1	6.7	1.7	17.5
U2 - IPDE alpha = 0.3	17.4	40.6	14.9	5	1	21.1
U2 - IPDE alpha = 0.6	20.3	40.3	11.4	3.5	0.5	24
U2 - IPDE alpha = 0.9	22.7	36.1	8.6	2.4	0.2	30
U3 - IPDE alpha = 0	14.9	43.9	17	5.3	1.3	17.6
U3 - IPDE alpha = 0.3	18.6	41.6	12.8	3.9	0.9	22.2
U3 - IPDE alpha = 0.6	21.9	39.7	10.6	2.5	0.4	24.9
U3 - IPDE alpha = 0.9	26.1	35.1	6.8	2	0.2	29.8
Mean patients treated						
U2 - IPDE alpha = 0	9.5	9.9	5	3	0.9	
U2 - IPDE alpha = 0.3	10.8	9.7	4.6	2.3	0.6	
U2 - IPDE alpha = 0.6	12	9.6	4.1	1.7	0.3	
U2 - IPDE alpha = 0.9	12.9	9.4	3.4	1.3	0.2	
U3 - IPDE alpha = 0	9.8	10.2	4.9	2.7	0.8	
U3 - IPDE alpha = 0.3	11	10	4.3	2.1	0.5	
U3 - IPDE alpha = 0.6	12.2	9.8	3.8	1.6	0.3	
U3 - IPDE alpha = 0.9	13.2	9.5	3.2	1.2	0.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.6	4.2	1.3	1	0.2	
U2 - IPDE alpha = 0.3	0.7	4.4	1.2	0.8	0.2	
U2 - IPDE alpha = 0.6	0.7	4.5	1.1	0.7	0.1	
U2 - IPDE alpha = 0.9	0.7	4.6	0.9	0.5	0.1	
U3 - IPDE alpha = 0	0.7	4.2	1.2	0.9	0.2	
U3 - IPDE alpha = 0.3	0.7	4.4	1.1	0.7	0.2	
U3 - IPDE alpha = 0.6	0.7	4.5	1	0.6	0.1	
U3 - IPDE alpha = 0.9	0.8	4.5	0.9	0.5	0.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - IPDE alpha = 0	18.6	17.6	13.7	34.4	14.9	0.8
U2 - IPDE alpha = 0.3	19.2	18.4	16.8	32.4	12.3	0.9
U2 - IPDE alpha = 0.6	19.6	21.2	19.6	28.5	9.6	1.5
U2 - IPDE alpha = 0.9	23.1	22.7	21.1	23.1	8.3	1.7
U3 - IPDE alpha = 0	26.8	18.9	11.6	30.2	11.6	0.9
U3 - IPDE alpha = 0.3	27.6	22.9	11.5	28.4	8.7	0.9
U3 - IPDE alpha = 0.6	27.6	24.1	14.1	25.4	7.5	1.3
U3 - IPDE alpha = 0.9	30.8	26.5	14.2	20.6	6.4	1.5
Mean patients treated						
U2 - IPDE alpha = 0	6.2	6.1	5.6	7.7	4.3	
U2 - IPDE alpha = 0.3	6.4	6.6	5.9	7.2	3.7	
U2 - IPDE alpha = 0.6	6.7	7	6.3	6.8	3.1	
U2 - IPDE alpha = 0.9	7.1	7.4	6.4	6.5	2.4	
U3 - IPDE alpha = 0	6.8	6	5.4	7.7	4	
U3 - IPDE alpha = 0.3	7.2	6.5	5.6	7.2	3.5	
U3 - IPDE alpha = 0.6	7.3	7	5.9	6.7	2.9	
U3 - IPDE alpha = 0.9	7.8	7.4	6.1	6.3	2.3	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.5	3.5	1.6	2.2	1	
U2 - IPDE alpha = 0.3	0.6	3.7	1.7	2.1	0.9	
U2 - IPDE alpha = 0.6	0.6	3.8	1.8	2.1	0.8	
U2 - IPDE alpha = 0.9	0.7	4	1.8	2	0.7	
U3 - IPDE alpha = 0	0.6	3.5	1.6	2.1	0.9	
U3 - IPDE alpha = 0.3	0.7	3.7	1.7	2.1	0.8	
U3 - IPDE alpha = 0.6	0.7	3.9	1.7	2	0.7	
U3 - IPDE alpha = 0.9	0.8	4	1.8	1.9	0.6	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - IPDE alpha = 0	47.7	30.2	13.8	6.1	0.6	1.6
U2 - IPDE alpha = 0.3	48.2	32.3	12.5	4.4	0.3	2.3
U2 - IPDE alpha = 0.6	46.8	35.2	12.3	3	0.1	2.6
U2 - IPDE alpha = 0.9	47.7	36.7	10.2	2.1	0.1	3.2
U3 - IPDE alpha = 0	61.3	26.2	8	2.5	0.4	1.6
U3 - IPDE alpha = 0.3	54.8	30.4	9.4	3.1	0.1	2.2
U3 - IPDE alpha = 0.6	54.3	32.3	8.5	2.2	0.1	2.6
U3 - IPDE alpha = 0.9	52.8	34.3	7.6	1.5	0	3.8
Mean patients treated						
U2 - IPDE alpha = 0	10.1	8.2	5.7	4.2	1.6	
U2 - IPDE alpha = 0.3	10.6	8.7	5.7	3.7	1.2	
U2 - IPDE alpha = 0.6	11.2	9.2	5.4	3.1	0.8	
U2 - IPDE alpha = 0.9	11.8	9.6	5	2.7	0.6	
U3 - IPDE alpha = 0	11.1	7.9	5.4	4	1.5	
U3 - IPDE alpha = 0.3	11.4	8.4	5.4	3.5	1.1	
U3 - IPDE alpha = 0.6	12.1	8.8	5.1	3	0.8	
U3 - IPDE alpha = 0.9	12.7	9.2	4.6	2.6	0.6	
Mean IPDE patients						
U2 - IPDE alpha = 0	1	4.1	1.8	1.6	0.4	
U2 - IPDE alpha = 0.3	1.1	4.2	1.8	1.4	0.4	
U2 - IPDE alpha = 0.6	1.1	4.4	1.8	1.3	0.3	
U2 - IPDE alpha = 0.9	1.1	4.5	1.7	1.1	0.2	
U3 - IPDE alpha = 0	1.2	4	1.7	1.5	0.4	
U3 - IPDE alpha = 0.3	1.2	4.1	1.7	1.4	0.4	
U3 - IPDE alpha = 0.6	1.2	4.2	1.7	1.2	0.3	
U3 - IPDE alpha = 0.9	1.3	4.4	1.6	1.1	0.2	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - IPDE alpha = 0	41.3	25	13.3	13.7	6.7	0
U2 - IPDE alpha = 0.3	38.1	26.9	15.2	12.2	7.6	0
U2 - IPDE alpha = 0.6	36	27.9	14.9	13.9	7.3	0
U2 - IPDE alpha = 0.9	31.5	30.6	16.5	13.6	7.8	0
U3 - IPDE alpha = 0	46.9	24.5	13.3	11.2	4.1	0
U3 - IPDE alpha = 0.3	44	27.4	14.7	9.8	4.1	0
U3 - IPDE alpha = 0.6	41.2	30.4	14.1	10	4.3	0
U3 - IPDE alpha = 0.9	37.1	32.4	14.1	10.8	5.6	0
Mean patients treated						
U2 - IPDE alpha = 0	4.5	4.9	6	8.3	6.4	
U2 - IPDE alpha = 0.3	4.3	4.9	5.9	8.3	6.5	
U2 - IPDE alpha = 0.6	4.3	5.3	6.1	8.2	6.1	
U2 - IPDE alpha = 0.9	4.2	5.6	6.3	8.2	5.7	
U3 - IPDE alpha = 0	4.7	5.1	6.2	8.3	5.8	
U3 - IPDE alpha = 0.3	4.5	5.3	6.2	8.2	5.8	
U3 - IPDE alpha = 0.6	4.6	5.7	6.2	8	5.5	
U3 - IPDE alpha = 0.9	4.6	6.2	6.4	7.8	5.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.2	2.8	2	2.9	1.4	
U2 - IPDE alpha = 0.3	0.2	2.7	2.1	2.9	1.5	
U2 - IPDE alpha = 0.6	0.2	2.8	2.2	2.9	1.5	
U2 - IPDE alpha = 0.9	0.2	2.8	2.3	2.9	1.6	
U3 - IPDE alpha = 0	0.2	2.8	2.1	2.9	1.4	
U3 - IPDE alpha = 0.3	0.2	2.8	2.2	2.8	1.4	
U3 - IPDE alpha = 0.6	0.2	2.9	2.2	2.9	1.4	
U3 - IPDE alpha = 0.9	0.3	2.9	2.4	2.7	1.4	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - IPDE alpha = 0	20.8	58.9	17.3	2.8	0.1	0.1
U2 - IPDE alpha = 0.3	18.7	62.6	17	1.5	0	0.2
U2 - IPDE alpha = 0.6	18.7	66.5	14	0.7	0	0.1
U2 - IPDE alpha = 0.9	18	69.3	12.4	0.2	0	0.1
U3 - IPDE alpha = 0	24.4	60	13.7	1.8	0	0.1
U3 - IPDE alpha = 0.3	22.1	64.7	12.4	0.7	0	0.1
U3 - IPDE alpha = 0.6	21.6	67.9	10.1	0.3	0	0.1
U3 - IPDE alpha = 0.9	22.3	68.7	8.7	0.2	0	0.1
Mean patients treated						
U2 - IPDE alpha = 0	9.1	14.1	5.3	1.3	0.2	
U2 - IPDE alpha = 0.3	9.4	15	4.6	0.8	0.1	
U2 - IPDE alpha = 0.6	9.6	15.9	4.1	0.4	0	
U2 - IPDE alpha = 0.9	10.2	16.3	3.3	0.2	0	
U3 - IPDE alpha = 0	9.8	14.4	4.7	1	0.1	
U3 - IPDE alpha = 0.3	9.9	15.5	3.9	0.6	0.1	
U3 - IPDE alpha = 0.6	10.3	16.2	3.2	0.3	0	
U3 - IPDE alpha = 0.9	10.8	16.5	2.5	0.2	0	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.5	3.7	1.2	0.4	0.1	
U2 - IPDE alpha = 0.3	0.5	3.9	1.2	0.3	0	
U2 - IPDE alpha = 0.6	0.5	4.1	1.2	0.2	0	
U2 - IPDE alpha = 0.9	0.6	4.2	1	0.1	0	
U3 - IPDE alpha = 0	0.5	3.7	1	0.3	0	
U3 - IPDE alpha = 0.3	0.6	3.9	1	0.2	0	
U3 - IPDE alpha = 0.6	0.7	4	0.8	0.1	0	
U3 - IPDE alpha = 0.9	0.7	4.2	0.7	0.1	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - IPDE alpha = 0	11.6	11.3	7.5	14.5	55.1	0
U2 - IPDE alpha = 0.3	9.7	11.4	8.5	16.7	53.7	0
U2 - IPDE alpha = 0.6	9.7	11.6	10.6	18.4	49.7	0
U2 - IPDE alpha = 0.9	9.4	11.7	9.6	21.7	47.6	0
U3 - IPDE alpha = 0	17.1	13.5	9.1	12.3	48	0
U3 - IPDE alpha = 0.3	17	14	9.7	13.6	45.7	0
U3 - IPDE alpha = 0.6	16.1	15.2	9.5	16.1	43.1	0
U3 - IPDE alpha = 0.9	14.9	15.3	10.1	18.6	41.1	0
Mean patients treated						
U2 - IPDE alpha = 0	3.5	4	5.1	6.8	10.6	
U2 - IPDE alpha = 0.3	3.5	4.1	5.2	7	10.2	
U2 - IPDE alpha = 0.6	3.4	4.1	5.2	7.3	9.9	
U2 - IPDE alpha = 0.9	3.4	4.2	5.3	7.6	9.4	
U3 - IPDE alpha = 0	3.6	4.1	5.2	6.9	10.2	
U3 - IPDE alpha = 0.3	3.6	4.2	5.4	7.1	9.7	
U3 - IPDE alpha = 0.6	3.5	4.2	5.4	7.5	9.4	
U3 - IPDE alpha = 0.9	3.6	4.4	5.5	7.7	8.8	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.1	2.7	1.5	2.5	1.5	
U2 - IPDE alpha = 0.3	0.1	2.7	1.5	2.5	1.6	
U2 - IPDE alpha = 0.6	0.1	2.7	1.5	2.5	1.7	
U2 - IPDE alpha = 0.9	0.1	2.8	1.5	2.5	1.8	
U3 - IPDE alpha = 0	0.2	2.8	1.5	2.5	1.5	
U3 - IPDE alpha = 0.3	0.2	2.8	1.5	2.5	1.6	
U3 - IPDE alpha = 0.6	0.1	2.8	1.6	2.5	1.6	
U3 - IPDE alpha = 0.9	0.2	2.8	1.6	2.6	1.7	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - IPDE alpha = 0	14.5	59	20.7	3.1	0.1	2.6
U2 - IPDE alpha = 0.3	16.6	59.4	19.4	1.4	0	3.2
U2 - IPDE alpha = 0.6	18.2	60	16.7	0.7	0	4.4
U2 - IPDE alpha = 0.9	22.6	60	12.6	0.4	0	4.4
U3 - IPDE alpha = 0	17.7	61.1	16	2.5	0.1	2.6
U3 - IPDE alpha = 0.3	20	59.6	15.9	1.3	0	3.2
U3 - IPDE alpha = 0.6	22.2	58.8	14.1	0.5	0	4.4
U3 - IPDE alpha = 0.9	24.6	59.6	10.2	0.3	0	5.3
Mean patients treated						
U2 - IPDE alpha = 0	8	12.6	6.6	2.3	0.4	
U2 - IPDE alpha = 0.3	8.5	13	6.2	1.9	0.2	
U2 - IPDE alpha = 0.6	9.3	13.2	5.6	1.5	0.1	
U2 - IPDE alpha = 0.9	9.9	13.5	5.1	1.2	0.1	
U3 - IPDE alpha = 0	8.4	12.8	6.2	2.1	0.3	
U3 - IPDE alpha = 0.3	8.9	13.2	5.9	1.7	0.2	
U3 - IPDE alpha = 0.6	9.6	13.4	5.3	1.4	0.1	
U3 - IPDE alpha = 0.9	10.3	13.4	4.8	1.1	0.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.7	4.6	1.7	0.9	0.1	
U2 - IPDE alpha = 0.3	0.7	4.8	1.7	0.8	0.1	
U2 - IPDE alpha = 0.6	0.7	4.9	1.7	0.7	0	
U2 - IPDE alpha = 0.9	0.8	5.3	1.6	0.6	0	
U3 - IPDE alpha = 0	0.7	4.6	1.6	0.8	0.1	
U3 - IPDE alpha = 0.3	0.7	4.7	1.6	0.7	0.1	
U3 - IPDE alpha = 0.6	0.8	4.9	1.5	0.6	0	
U3 - IPDE alpha = 0.9	0.9	5.1	1.5	0.5	0	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - IPDE alpha = 0	9.8	48.5	27.5	12.1	1.8	0.3
U2 - IPDE alpha = 0.3	9.1	50.3	29.1	10.3	0.8	0.4
U2 - IPDE alpha = 0.6	9.2	50.5	30.2	8.9	0.7	0.5
U2 - IPDE alpha = 0.9	10.6	52.6	28.4	7.5	0.2	0.7
U3 - IPDE alpha = 0	14	50.4	23.6	10.4	1.3	0.3
U3 - IPDE alpha = 0.3	13.9	53.2	23.1	8.7	0.7	0.4
U3 - IPDE alpha = 0.6	13.7	53.8	24.1	7.5	0.4	0.5
U3 - IPDE alpha = 0.9	13.6	55.6	24.2	5.8	0.2	0.6
Mean patients treated						
U2 - IPDE alpha = 0	4.8	10	8.2	5.2	1.9	
U2 - IPDE alpha = 0.3	5	10.6	8.3	4.7	1.4	
U2 - IPDE alpha = 0.6	5.2	11	8.4	4.2	1.1	
U2 - IPDE alpha = 0.9	5.4	11.6	8.4	3.7	0.8	
U3 - IPDE alpha = 0	5.3	10.4	7.8	4.8	1.6	
U3 - IPDE alpha = 0.3	5.4	11	7.8	4.4	1.3	
U3 - IPDE alpha = 0.6	5.6	11.4	7.9	4	1	
U3 - IPDE alpha = 0.9	5.8	11.9	8	3.5	0.7	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.4	3.9	2.3	1.7	0.6	
U2 - IPDE alpha = 0.3	0.4	3.9	2.3	1.6	0.4	
U2 - IPDE alpha = 0.6	0.4	4.1	2.4	1.5	0.4	
U2 - IPDE alpha = 0.9	0.4	4.2	2.4	1.4	0.3	
U3 - IPDE alpha = 0	0.5	3.8	2.2	1.6	0.5	
U3 - IPDE alpha = 0.3	0.5	4	2.2	1.5	0.4	
U3 - IPDE alpha = 0.6	0.5	4	2.3	1.5	0.3	
U3 - IPDE alpha = 0.9	0.5	4.2	2.3	1.4	0.3	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - IPDE alpha = 0	57.1	1.8	0	0	0	41.1
U2 - IPDE alpha = 0.3	55.8	1.4	0	0	0	42.8
U2 - IPDE alpha = 0.6	54	1.2	0	0	0	44.8
U2 - IPDE alpha = 0.9	52	0.5	0	0	0	47.5
U3 - IPDE alpha = 0	57.9	1.5	0	0	0	40.6
U3 - IPDE alpha = 0.3	56.5	1.2	0	0	0	42.3
U3 - IPDE alpha = 0.6	54.4	0.8	0	0	0	44.8
U3 - IPDE alpha = 0.9	52.3	0.5	0	0	0	47.2
Mean patients treated						
U2 - IPDE alpha = 0	20.2	2.5	0.2	0	0	
U2 - IPDE alpha = 0.3	20.1	2.4	0.1	0	0	
U2 - IPDE alpha = 0.6	19.9	2.3	0	0	0	
U2 - IPDE alpha = 0.9	19.7	2.1	0	0	0	
U3 - IPDE alpha = 0	20.4	2.5	0.2	0	0	
U3 - IPDE alpha = 0.3	20.2	2.4	0.1	0	0	
U3 - IPDE alpha = 0.6	20	2.2	0	0	0	
U3 - IPDE alpha = 0.9	19.8	2.1	0	0	0	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.2	1.1	0.1	0	0	
U2 - IPDE alpha = 0.3	0.2	1.1	0.1	0	0	
U2 - IPDE alpha = 0.6	0.2	1.1	0	0	0	
U2 - IPDE alpha = 0.9	0.2	1.1	0	0	0	
U3 - IPDE alpha = 0	0.2	1.1	0.1	0	0	
U3 - IPDE alpha = 0.3	0.2	1.1	0	0	0	
U3 - IPDE alpha = 0.6	0.2	1.1	0	0	0	
U3 - IPDE alpha = 0.9	0.2	1.1	0	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - IPDE alpha = 0	18.4	62.5	14.1	3.7	0	1.3
U2 - IPDE alpha = 0.3	19.5	64.1	12.7	1.8	0	1.9
U2 - IPDE alpha = 0.6	21.6	64.9	10	1.4	0	2.1
U2 - IPDE alpha = 0.9	23.7	64.5	8.9	0.8	0	2.1
U3 - IPDE alpha = 0	23	60.5	11.4	3.5	0	1.6
U3 - IPDE alpha = 0.3	22.2	62	11.8	1.8	0	2.2
U3 - IPDE alpha = 0.6	25.3	61.1	9.6	1.7	0	2.3
U3 - IPDE alpha = 0.9	26.2	61.5	9.2	0.9	0	2.2
Mean patients treated						
U2 - IPDE alpha = 0	8.5	13.3	5.7	2.2	0.2	
U2 - IPDE alpha = 0.3	9	13.5	5.4	1.9	0.1	
U2 - IPDE alpha = 0.6	9.4	13.9	5.1	1.5	0.1	
U2 - IPDE alpha = 0.9	10	14	4.6	1.2	0	
U3 - IPDE alpha = 0	9.2	13.1	5.4	2.1	0.2	
U3 - IPDE alpha = 0.3	9.5	13.5	5	1.7	0.1	
U3 - IPDE alpha = 0.6	9.8	13.8	4.8	1.4	0.1	
U3 - IPDE alpha = 0.9	10.5	13.8	4.4	1.1	0	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.9	4.8	1.7	0.7	0.1	
U2 - IPDE alpha = 0.3	1	4.9	1.7	0.6	0	
U2 - IPDE alpha = 0.6	1	5	1.6	0.6	0	
U2 - IPDE alpha = 0.9	1	5.3	1.6	0.5	0	
U3 - IPDE alpha = 0	1	4.7	1.6	0.7	0.1	
U3 - IPDE alpha = 0.3	1	4.7	1.6	0.6	0	
U3 - IPDE alpha = 0.6	1.1	4.9	1.6	0.6	0	
U3 - IPDE alpha = 0.9	1.1	5.1	1.5	0.5	0	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - IPDE alpha = 0	21	4.2	0	0	0	74.8
U2 - IPDE alpha = 0.3	18.5	2.9	0	0	0	78.6
U2 - IPDE alpha = 0.6	17.5	2	0	0	0	80.5
U2 - IPDE alpha = 0.9	16.7	2.2	0	0	0	81.1
U3 - IPDE alpha = 0	20.9	3.9	0.1	0	0	75.1
U3 - IPDE alpha = 0.3	18.7	3.1	0	0	0	78.2
U3 - IPDE alpha = 0.6	17.7	1.7	0	0	0	80.6
U3 - IPDE alpha = 0.9	16.8	1.6	0	0	0	81.6
Mean patients treated						
U2 - IPDE alpha = 0	15.9	4.8	0.7	0.1	0	
U2 - IPDE alpha = 0.3	15.9	4.4	0.5	0	0	
U2 - IPDE alpha = 0.6	15.4	4	0.3	0	0	
U2 - IPDE alpha = 0.9	15.1	3.6	0.2	0	0	
U3 - IPDE alpha = 0	15.9	4.7	0.7	0.1	0	
U3 - IPDE alpha = 0.3	15.8	4.2	0.5	0	0	
U3 - IPDE alpha = 0.6	15.5	3.9	0.3	0	0	
U3 - IPDE alpha = 0.9	15.1	3.5	0.2	0	0	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.7	2.8	0.2	0	0	
U2 - IPDE alpha = 0.3	0.6	2.6	0.2	0	0	
U2 - IPDE alpha = 0.6	0.6	2.4	0.1	0	0	
U2 - IPDE alpha = 0.9	0.5	2.2	0.1	0	0	
U3 - IPDE alpha = 0	0.7	2.7	0.2	0	0	
U3 - IPDE alpha = 0.3	0.6	2.5	0.2	0	0	
U3 - IPDE alpha = 0.6	0.6	2.3	0.1	0	0	
U3 - IPDE alpha = 0.9	0.5	2.2	0.1	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - IPDE alpha = 0	2.3	1.4	3.1	56.3	36.9	0
U2 - IPDE alpha = 0.3	2.1	0.6	3.2	56.3	37.8	0
U2 - IPDE alpha = 0.6	1.3	0.8	3	53.4	41.5	0
U2 - IPDE alpha = 0.9	0.8	0.9	2.3	53.9	42.1	0
U3 - IPDE alpha = 0	3.9	3	4.8	58.6	29.7	0
U3 - IPDE alpha = 0.3	2.8	2.5	4	58.9	31.8	0
U3 - IPDE alpha = 0.6	2.3	2	3.7	57.2	34.8	0
U3 - IPDE alpha = 0.9	1.3	1.3	3.6	58.7	35.1	0
Mean patients treated						
U2 - IPDE alpha = 0	3.1	3.1	4	12.5	7.2	
U2 - IPDE alpha = 0.3	3.1	3.2	3.9	12.8	7	
U2 - IPDE alpha = 0.6	3.1	3.2	3.8	12.7	7.2	
U2 - IPDE alpha = 0.9	3.1	3.2	3.8	12.9	7	
U3 - IPDE alpha = 0	3.1	3.2	4.1	13.1	6.5	
U3 - IPDE alpha = 0.3	3.1	3.2	4	13.4	6.4	
U3 - IPDE alpha = 0.6	3.1	3.2	4	13.4	6.4	
U3 - IPDE alpha = 0.9	3.1	3.2	3.9	13.4	6.4	
Mean IPDE patients						
U2 - IPDE alpha = 0	0	2.4	1.1	2.5	1.4	
U2 - IPDE alpha = 0.3	0	2.5	1.1	2.4	1.4	
U2 - IPDE alpha = 0.6	0	2.4	1	2.4	1.5	
U2 - IPDE alpha = 0.9	0	2.4	1	2.4	1.5	
U3 - IPDE alpha = 0	0	2.5	1.1	2.5	1.2	
U3 - IPDE alpha = 0.3	0	2.4	1.1	2.4	1.3	
U3 - IPDE alpha = 0.6	0	2.4	1.1	2.4	1.3	
U3 - IPDE alpha = 0.9	0	2.4	1.1	2.5	1.3	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - IPDE alpha = 0	28.1	28.6	37	6.1	0.1	0.1
U2 - IPDE alpha = 0.3	27.4	33.6	34.4	4.2	0.1	0.3
U2 - IPDE alpha = 0.6	26.6	37.4	32.6	3.3	0	0.1
U2 - IPDE alpha = 0.9	28.5	40.9	29	1.5	0	0.1
U3 - IPDE alpha = 0	32.8	30.6	32	4.5	0	0.1
U3 - IPDE alpha = 0.3	33.4	33.9	29.4	3.2	0	0.1
U3 - IPDE alpha = 0.6	33.1	35.6	28.5	2.7	0	0.1
U3 - IPDE alpha = 0.9	32.7	39.2	26.2	1.7	0	0.2
Mean patients treated						
U2 - IPDE alpha = 0	8.9	9	8.8	2.8	0.4	
U2 - IPDE alpha = 0.3	9	9.9	8.4	2.5	0.2	
U2 - IPDE alpha = 0.6	9.1	10.5	8.2	2	0.1	
U2 - IPDE alpha = 0.9	9.1	11.2	8	1.5	0.1	
U3 - IPDE alpha = 0	9.6	9.2	8.3	2.5	0.3	
U3 - IPDE alpha = 0.3	9.5	9.9	8.2	2.3	0.2	
U3 - IPDE alpha = 0.6	9.7	10.6	7.8	1.8	0.1	
U3 - IPDE alpha = 0.9	9.8	11.2	7.6	1.4	0.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.7	3.7	1.9	1	0.1	
U2 - IPDE alpha = 0.3	0.7	3.7	2	0.9	0.1	
U2 - IPDE alpha = 0.6	0.8	4	2.1	0.8	0.1	
U2 - IPDE alpha = 0.9	0.8	4.1	2.1	0.7	0	
U3 - IPDE alpha = 0	0.8	3.6	1.8	0.9	0.1	
U3 - IPDE alpha = 0.3	0.8	3.7	1.9	0.9	0.1	
U3 - IPDE alpha = 0.6	0.9	4	1.9	0.8	0	
U3 - IPDE alpha = 0.9	0.9	4.1	2	0.7	0	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - IPDE alpha = 0	22.8	32.2	34.1	7.4	0.1	3.4
U2 - IPDE alpha = 0.3	23.4	35.9	31.3	5.4	0.1	3.9
U2 - IPDE alpha = 0.6	25.2	40.6	27.3	3	0	3.9
U2 - IPDE alpha = 0.9	27.5	43.4	21.8	1.9	0	5.4
U3 - IPDE alpha = 0	28.6	31.3	30.2	6.5	0.1	3.3
U3 - IPDE alpha = 0.3	28.4	35.8	27.5	4.8	0.1	3.4
U3 - IPDE alpha = 0.6	30.4	39.2	23.9	2.6	0	3.9
U3 - IPDE alpha = 0.9	31.8	41.3	19.2	2.1	0	5.6
Mean patients treated						
U2 - IPDE alpha = 0	8.8	8.9	8.4	3.2	0.4	
U2 - IPDE alpha = 0.3	9.3	9.5	8	2.7	0.3	
U2 - IPDE alpha = 0.6	10.1	10.1	7.1	2.2	0.2	
U2 - IPDE alpha = 0.9	10.7	10.4	6.5	1.8	0.1	
U3 - IPDE alpha = 0	9.4	8.8	8.1	2.9	0.4	
U3 - IPDE alpha = 0.3	9.9	9.5	7.6	2.5	0.2	
U3 - IPDE alpha = 0.6	10.6	10	6.8	2	0.1	
U3 - IPDE alpha = 0.9	11.4	10.2	6.2	1.7	0.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.7	4	1.7	1.3	0.1	
U2 - IPDE alpha = 0.3	0.7	4.2	1.7	1.1	0.1	
U2 - IPDE alpha = 0.6	0.8	4.4	1.6	1	0.1	
U2 - IPDE alpha = 0.9	0.8	4.6	1.6	0.9	0	
U3 - IPDE alpha = 0	0.8	3.9	1.6	1.2	0.1	
U3 - IPDE alpha = 0.3	0.8	4.2	1.5	1.1	0.1	
U3 - IPDE alpha = 0.6	0.8	4.3	1.6	0.9	0	
U3 - IPDE alpha = 0.9	0.9	4.5	1.5	0.8	0	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - IPDE alpha = 0	14.6	26.6	46.7	11	0.5	0.6
U2 - IPDE alpha = 0.3	15.8	25.9	46.9	9.9	0.4	1.1
U2 - IPDE alpha = 0.6	17.1	30.6	43.4	7.6	0	1.3
U2 - IPDE alpha = 0.9	17.5	34.7	41.3	5.1	0	1.4
U3 - IPDE alpha = 0	20.7	26.4	42.6	9.1	0.4	0.8
U3 - IPDE alpha = 0.3	20.7	28.3	42	7.6	0.3	1.1
U3 - IPDE alpha = 0.6	21	32.5	40	5.2	0.1	1.2
U3 - IPDE alpha = 0.9	21.3	35.4	37.6	4.3	0	1.4
Mean patients treated						
U2 - IPDE alpha = 0	6.6	8.2	10.2	4	0.9	
U2 - IPDE alpha = 0.3	6.9	8.7	10.2	3.6	0.6	
U2 - IPDE alpha = 0.6	7.3	9.3	9.7	3.2	0.4	
U2 - IPDE alpha = 0.9	7.8	9.8	9.3	2.8	0.2	
U3 - IPDE alpha = 0	7.2	8.1	10.1	3.8	0.7	
U3 - IPDE alpha = 0.3	7.4	8.8	9.8	3.4	0.5	
U3 - IPDE alpha = 0.6	7.8	9.3	9.4	3.1	0.3	
U3 - IPDE alpha = 0.9	8.3	9.8	9	2.6	0.2	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.5	3.7	2.1	1.4	0.2	
U2 - IPDE alpha = 0.3	0.5	3.8	2.2	1.4	0.2	
U2 - IPDE alpha = 0.6	0.6	4	2.1	1.3	0.1	
U2 - IPDE alpha = 0.9	0.6	4.2	2.1	1.1	0.1	
U3 - IPDE alpha = 0	0.6	3.7	2	1.4	0.2	
U3 - IPDE alpha = 0.3	0.7	3.8	2	1.3	0.2	
U3 - IPDE alpha = 0.6	0.7	3.9	2.1	1.2	0.1	
U3 - IPDE alpha = 0.9	0.7	4.2	2.1	1.1	0.1	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - IPDE alpha = 0	18.7	52	9.4	0.5	0	19.4
U2 - IPDE alpha = 0.3	22.9	47.5	6.5	0	0	23.1
U2 - IPDE alpha = 0.6	27.4	43.1	4.3	0.1	0	25.1
U2 - IPDE alpha = 0.9	29.4	37.2	2.8	0	0	30.6
U3 - IPDE alpha = 0	20.6	51.6	7.8	0.3	0	19.7
U3 - IPDE alpha = 0.3	23.5	46.7	5.9	0.1	0	23.8
U3 - IPDE alpha = 0.6	28.4	42.5	3.3	0	0	25.8
U3 - IPDE alpha = 0.9	30.7	35.6	2.7	0	0	31
Mean patients treated						
U2 - IPDE alpha = 0	10.7	12.1	4.1	1.2	0.2	
U2 - IPDE alpha = 0.3	12.1	11.6	3.4	0.9	0.1	
U2 - IPDE alpha = 0.6	13.1	11.2	2.8	0.6	0	
U2 - IPDE alpha = 0.9	14.1	10.4	2.4	0.5	0	
U3 - IPDE alpha = 0	11	12	3.9	1.1	0.1	
U3 - IPDE alpha = 0.3	12.3	11.6	3.2	0.8	0.1	
U3 - IPDE alpha = 0.6	13.4	11	2.7	0.6	0	
U3 - IPDE alpha = 0.9	14.4	10.3	2.2	0.4	0	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.8	4.6	1.2	0.4	0.1	
U2 - IPDE alpha = 0.3	0.9	4.6	1.1	0.3	0	
U2 - IPDE alpha = 0.6	0.9	4.8	1	0.3	0	
U2 - IPDE alpha = 0.9	0.9	4.8	0.8	0.2	0	
U3 - IPDE alpha = 0	0.9	4.5	1.2	0.4	0	
U3 - IPDE alpha = 0.3	0.9	4.6	1	0.3	0	
U3 - IPDE alpha = 0.6	0.9	4.6	0.9	0.3	0	
U3 - IPDE alpha = 0.9	0.9	4.7	0.8	0.2	0	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - IPDE alpha = 0	66.1	27.1	4.2	0.2	0	2.4
U2 - IPDE alpha = 0.3	62.2	30.7	4.6	0.1	0	2.4
U2 - IPDE alpha = 0.6	62.7	32	2.6	0.1	0	2.6
U2 - IPDE alpha = 0.9	60.7	34.7	2	0	0	2.6
U3 - IPDE alpha = 0	69.6	25.2	2.8	0.1	0	2.3
U3 - IPDE alpha = 0.3	65.3	30.1	2.3	0.1	0	2.2
U3 - IPDE alpha = 0.6	65.7	30.1	1.8	0	0	2.4
U3 - IPDE alpha = 0.9	64.4	31.9	1	0	0	2.7
Mean patients treated						
U2 - IPDE alpha = 0	17.7	8.5	2.7	0.6	0	
U2 - IPDE alpha = 0.3	17.9	9.1	2.2	0.4	0	
U2 - IPDE alpha = 0.6	18.3	9.2	1.7	0.2	0	
U2 - IPDE alpha = 0.9	18.3	9.7	1.3	0.1	0	
U3 - IPDE alpha = 0	18.7	8.2	2.2	0.5	0	
U3 - IPDE alpha = 0.3	18.8	8.7	1.7	0.3	0	
U3 - IPDE alpha = 0.6	19.1	9	1.3	0.2	0	
U3 - IPDE alpha = 0.9	19	9.5	0.9	0.1	0	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.8	3	0.9	0.2	0	
U2 - IPDE alpha = 0.3	0.8	3.2	0.8	0.2	0	
U2 - IPDE alpha = 0.6	0.9	3.3	0.8	0.1	0	
U2 - IPDE alpha = 0.9	0.9	3.6	0.7	0	0	
U3 - IPDE alpha = 0	0.8	2.9	0.7	0.2	0	
U3 - IPDE alpha = 0.3	0.9	3.1	0.6	0.1	0	
U3 - IPDE alpha = 0.6	0.9	3.1	0.5	0.1	0	
U3 - IPDE alpha = 0.9	1	3.4	0.4	0	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - IPDE alpha = 0	17.2	11.9	11.2	45.3	14.4	0
U2 - IPDE alpha = 0.3	17.3	11.8	14.7	44	12.2	0
U2 - IPDE alpha = 0.6	15.9	14.2	15.5	42.8	11.6	0
U2 - IPDE alpha = 0.9	15	17	18.5	39.7	9.8	0
U3 - IPDE alpha = 0	27.9	14	9.8	38.3	10	0
U3 - IPDE alpha = 0.3	24.2	15	12.9	37.1	10.8	0
U3 - IPDE alpha = 0.6	22.8	16.8	14	37.4	9	0
U3 - IPDE alpha = 0.9	20.1	19.6	17.1	35.9	7.3	0
Mean patients treated						
U2 - IPDE alpha = 0	4.3	4.8	6	10.3	4.6	
U2 - IPDE alpha = 0.3	4.4	5.1	6.4	10.1	4	
U2 - IPDE alpha = 0.6	4.6	5.5	6.6	9.9	3.4	
U2 - IPDE alpha = 0.9	4.7	6.1	6.8	9.5	3	
U3 - IPDE alpha = 0	4.6	4.9	5.9	10.3	4.2	
U3 - IPDE alpha = 0.3	4.8	5.3	6.4	10	3.6	
U3 - IPDE alpha = 0.6	4.9	5.6	6.6	9.8	3.1	
U3 - IPDE alpha = 0.9	5	6.3	6.9	9.4	2.5	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.2	2.8	1.7	2.4	1.2	
U2 - IPDE alpha = 0.3	0.2	2.9	1.8	2.4	1.2	
U2 - IPDE alpha = 0.6	0.3	2.9	1.9	2.3	1.1	
U2 - IPDE alpha = 0.9	0.3	3	2	2.3	1	
U3 - IPDE alpha = 0	0.3	2.8	1.7	2.4	1.1	
U3 - IPDE alpha = 0.3	0.3	2.9	1.8	2.4	1	
U3 - IPDE alpha = 0.6	0.3	3	2	2.3	1	
U3 - IPDE alpha = 0.9	0.4	3.1	2.1	2.2	0.9	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - IPDE alpha = 0	4.9	35.7	26.7	26.2	6.5	0
U2 - IPDE alpha = 0.3	3.7	36.4	30.5	23.5	5.9	0
U2 - IPDE alpha = 0.6	2.8	40.2	29.2	24.5	3.3	0
U2 - IPDE alpha = 0.9	2.6	43.8	34	18	1.6	0
U3 - IPDE alpha = 0	7.7	42.5	25.4	20.2	4.2	0
U3 - IPDE alpha = 0.3	5.7	41.3	29.5	20.5	3	0
U3 - IPDE alpha = 0.6	3.9	46.9	28.1	19.6	1.5	0
U3 - IPDE alpha = 0.9	3.1	48.9	31.9	15	1.1	0
Mean patients treated						
U2 - IPDE alpha = 0	3.8	11.1	7.2	5.8	2.1	
U2 - IPDE alpha = 0.3	3.9	12.1	7.6	5	1.4	
U2 - IPDE alpha = 0.6	3.9	13.3	7.7	4.4	0.6	
U2 - IPDE alpha = 0.9	4	14.5	8.2	3.1	0.3	
U3 - IPDE alpha = 0	3.9	12	7.1	5.3	1.6	
U3 - IPDE alpha = 0.3	3.9	12.9	7.5	4.6	1	
U3 - IPDE alpha = 0.6	4	14.2	7.5	3.9	0.5	
U3 - IPDE alpha = 0.9	4.1	15.3	7.7	2.7	0.2	
Mean IPDE patients						
U2 - IPDE alpha = 0	0	2.6	1.5	1.3	0.6	
U2 - IPDE alpha = 0.3	0.1	2.8	1.6	1.1	0.5	
U2 - IPDE alpha = 0.6	0.1	2.8	1.7	1.1	0.3	
U2 - IPDE alpha = 0.9	0.1	2.9	1.8	0.8	0.2	
U3 - IPDE alpha = 0	0.1	2.7	1.4	1.1	0.5	
U3 - IPDE alpha = 0.3	0.1	2.7	1.5	1	0.3	
U3 - IPDE alpha = 0.6	0.1	2.9	1.6	0.9	0.2	
U3 - IPDE alpha = 0.9	0.1	2.9	1.6	0.7	0.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - IPDE alpha = 0	13.6	15.1	33.3	26.2	11.8	0
U2 - IPDE alpha = 0.3	11.7	14.8	33.1	28.8	11.6	0
U2 - IPDE alpha = 0.6	11.4	15.4	31.2	32.2	9.8	0
U2 - IPDE alpha = 0.9	8.8	14.3	33	33.9	10	0
U3 - IPDE alpha = 0	20	16.3	33.3	23.4	7	0
U3 - IPDE alpha = 0.3	17.5	16.7	31.7	27.5	6.6	0
U3 - IPDE alpha = 0.6	16.4	19.1	32	26.5	6	0
U3 - IPDE alpha = 0.9	14	18.5	31.2	30.6	5.7	0
Mean patients treated						
U2 - IPDE alpha = 0	3.5	4.3	8.5	8.5	5.3	
U2 - IPDE alpha = 0.3	3.5	4.3	8.6	8.7	4.9	
U2 - IPDE alpha = 0.6	3.5	4.5	8.6	9	4.4	
U2 - IPDE alpha = 0.9	3.5	4.6	8.7	9.1	4.1	
U3 - IPDE alpha = 0	3.6	4.5	8.9	8.4	4.6	
U3 - IPDE alpha = 0.3	3.6	4.6	9.1	8.6	4.1	
U3 - IPDE alpha = 0.6	3.6	4.7	9.1	8.8	3.7	
U3 - IPDE alpha = 0.9	3.6	5	9	9	3.4	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.1	2.7	1.9	2.8	1.1	
U2 - IPDE alpha = 0.3	0.1	2.7	1.9	2.8	1.2	
U2 - IPDE alpha = 0.6	0.1	2.7	1.9	2.9	1.2	
U2 - IPDE alpha = 0.9	0.1	2.8	2	2.9	1.1	
U3 - IPDE alpha = 0	0.1	2.8	1.9	2.7	1	
U3 - IPDE alpha = 0.3	0.2	2.8	2	2.8	0.9	
U3 - IPDE alpha = 0.6	0.2	2.8	2.1	2.8	1	
U3 - IPDE alpha = 0.9	0.1	2.9	2.1	2.8	1	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - IPDE alpha = 0	8.3	5.1	10.4	45.1	31.1	0
U2 - IPDE alpha = 0.3	8.4	5.4	11.5	46.8	27.9	0
U2 - IPDE alpha = 0.6	6.8	6.8	11.4	48.8	26.2	0
U2 - IPDE alpha = 0.9	7.1	8	14.6	48.1	22.2	0
U3 - IPDE alpha = 0	12.6	7.5	12.7	43	24.2	0
U3 - IPDE alpha = 0.3	12.6	7.7	12.9	44	22.8	0
U3 - IPDE alpha = 0.6	12.2	7.3	12.2	46.2	22.1	0
U3 - IPDE alpha = 0.9	11.4	8.8	13.2	48.7	17.9	0
Mean patients treated						
U2 - IPDE alpha = 0	3.3	3.6	5.9	10.3	6.8	
U2 - IPDE alpha = 0.3	3.3	3.7	6.1	10.6	6.2	
U2 - IPDE alpha = 0.6	3.3	3.8	6.1	10.8	5.9	
U2 - IPDE alpha = 0.9	3.3	3.9	6.4	11.1	5.3	
U3 - IPDE alpha = 0	3.5	3.7	6	10.5	6.3	
U3 - IPDE alpha = 0.3	3.5	3.9	6.2	10.7	5.8	
U3 - IPDE alpha = 0.6	3.4	4	6.3	11	5.4	
U3 - IPDE alpha = 0.9	3.4	4	6.5	11.2	4.9	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.1	2.6	1.5	2.8	1.2	
U2 - IPDE alpha = 0.3	0.1	2.7	1.5	2.7	1.2	
U2 - IPDE alpha = 0.6	0.1	2.7	1.5	2.8	1.2	
U2 - IPDE alpha = 0.9	0.1	2.7	1.6	2.9	1.2	
U3 - IPDE alpha = 0	0.1	2.7	1.5	2.7	1.1	
U3 - IPDE alpha = 0.3	0.1	2.7	1.5	2.7	1.1	
U3 - IPDE alpha = 0.6	0.1	2.7	1.6	2.8	1.1	
U3 - IPDE alpha = 0.9	0.1	2.7	1.6	2.8	1.1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - IPDE alpha = 0	14.4	12.8	44.8	22.3	5.5	0.2
U2 - IPDE alpha = 0.3	14.2	15.4	46.7	20.5	3	0.2
U2 - IPDE alpha = 0.6	13	17.8	49.2	17.1	2.6	0.3
U2 - IPDE alpha = 0.9	14.9	22.2	46.2	14.2	2.2	0.3
U3 - IPDE alpha = 0	17.9	15.2	43	20.4	3.3	0.2
U3 - IPDE alpha = 0.3	18.5	16.4	42.4	20.1	2.3	0.3
U3 - IPDE alpha = 0.6	17.2	19.8	45.3	15.8	1.7	0.2
U3 - IPDE alpha = 0.9	19.9	21.7	43.5	13.1	1.4	0.4
Mean patients treated						
U2 - IPDE alpha = 0	5.1	6.3	9.6	6.2	2.8	
U2 - IPDE alpha = 0.3	5.4	6.8	9.8	5.9	2.2	
U2 - IPDE alpha = 0.6	5.6	7.1	10	5.7	1.6	
U2 - IPDE alpha = 0.9	5.8	7.6	10	5.2	1.3	
U3 - IPDE alpha = 0	5.8	6.3	9.5	6	2.4	
U3 - IPDE alpha = 0.3	6	6.7	9.7	5.7	1.8	
U3 - IPDE alpha = 0.6	6.1	7.1	9.9	5.5	1.4	
U3 - IPDE alpha = 0.9	6.4	7.6	9.8	5.1	1.1	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.4	3.4	2.1	2	0.7	
U2 - IPDE alpha = 0.3	0.5	3.6	2.1	1.9	0.6	
U2 - IPDE alpha = 0.6	0.6	3.7	2.3	1.9	0.5	
U2 - IPDE alpha = 0.9	0.6	3.8	2.3	1.8	0.5	
U3 - IPDE alpha = 0	0.6	3.5	2	1.9	0.6	
U3 - IPDE alpha = 0.3	0.6	3.6	2.1	1.8	0.5	
U3 - IPDE alpha = 0.6	0.6	3.6	2.2	1.8	0.5	
U3 - IPDE alpha = 0.9	0.7	3.7	2.2	1.7	0.4	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - IPDE alpha = 0	10.2	40.3	31.5	14.3	3.7	0
U2 - IPDE alpha = 0.3	9.6	41.1	33.6	13.4	2.3	0
U2 - IPDE alpha = 0.6	7.8	44.5	35.2	11.1	1.4	0
U2 - IPDE alpha = 0.9	6.7	47.1	37.9	7.5	0.8	0
U3 - IPDE alpha = 0	13.8	46.6	27.7	9.4	2.5	0
U3 - IPDE alpha = 0.3	13.7	46.7	28.4	10	1.2	0
U3 - IPDE alpha = 0.6	10.7	48.4	32	8.2	0.7	0
U3 - IPDE alpha = 0.9	9.5	52.1	32.2	5.8	0.4	0
Mean patients treated						
U2 - IPDE alpha = 0	4.4	11.2	8.5	4.4	1.6	
U2 - IPDE alpha = 0.3	4.4	12.1	8.5	3.9	1	
U2 - IPDE alpha = 0.6	4.4	12.9	8.9	3.2	0.6	
U2 - IPDE alpha = 0.9	4.5	13.9	9.1	2.3	0.3	
U3 - IPDE alpha = 0	4.6	12.2	7.9	4	1.3	
U3 - IPDE alpha = 0.3	4.6	12.9	8.1	3.6	0.8	
U3 - IPDE alpha = 0.6	4.6	13.7	8.3	2.9	0.5	
U3 - IPDE alpha = 0.9	4.7	14.9	8.2	2.1	0.2	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.2	3	1.9	1.2	0.4	
U2 - IPDE alpha = 0.3	0.2	3.1	2	1.1	0.3	
U2 - IPDE alpha = 0.6	0.2	3.2	2.1	1	0.2	
U2 - IPDE alpha = 0.9	0.2	3.3	2.3	0.8	0.1	
U3 - IPDE alpha = 0	0.2	3	1.8	1.1	0.3	
U3 - IPDE alpha = 0.3	0.2	3.1	1.9	1	0.2	
U3 - IPDE alpha = 0.6	0.3	3.1	2	0.9	0.1	
U3 - IPDE alpha = 0.9	0.2	3.2	2	0.7	0.1	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - IPDE alpha = 0	20.8	48.7	15.4	7.8	1.6	5.7
U2 - IPDE alpha = 0.3	21.3	46.9	17.7	4.5	1.2	8.4
U2 - IPDE alpha = 0.6	24	48.5	12.7	3.9	0.8	10.1
U2 - IPDE alpha = 0.9	26.4	46.8	11.7	2.4	0.6	12.1
U3 - IPDE alpha = 0	25	47.6	13	7.2	1.2	6
U3 - IPDE alpha = 0.3	26	47.6	13.2	4.3	0.7	8.2
U3 - IPDE alpha = 0.6	27.3	48	10.7	3.6	0.4	10
U3 - IPDE alpha = 0.9	29	46.9	8.4	2.3	0.4	13
Mean patients treated						
U2 - IPDE alpha = 0	8.7	10.5	5.5	3.5	1.2	
U2 - IPDE alpha = 0.3	9.3	10.9	5.4	2.9	0.8	
U2 - IPDE alpha = 0.6	10.2	11.1	4.9	2.5	0.6	
U2 - IPDE alpha = 0.9	11.1	11.1	4.6	2	0.3	
U3 - IPDE alpha = 0	9.3	10.3	5.4	3.4	1.1	
U3 - IPDE alpha = 0.3	9.9	10.7	5.1	2.8	0.7	
U3 - IPDE alpha = 0.6	10.7	11	4.6	2.4	0.5	
U3 - IPDE alpha = 0.9	11.6	11	4.3	1.9	0.3	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.7	4.6	1.6	1.1	0.3	
U2 - IPDE alpha = 0.3	0.8	4.9	1.6	1	0.2	
U2 - IPDE alpha = 0.6	0.8	5.1	1.4	0.9	0.2	
U2 - IPDE alpha = 0.9	0.9	5.2	1.4	0.8	0.1	
U3 - IPDE alpha = 0	0.8	4.6	1.5	1.1	0.3	
U3 - IPDE alpha = 0.3	0.9	4.9	1.5	1	0.2	
U3 - IPDE alpha = 0.6	0.9	5	1.3	0.9	0.2	
U3 - IPDE alpha = 0.9	0.9	5.1	1.3	0.8	0.1	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = one_stage - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - IPDE alpha = 0	18.9	57.1	16.3	3.8	0.5	3.4
U2 - IPDE alpha = 0.3	20.3	58.3	14.6	2.6	0.3	3.9
U2 - IPDE alpha = 0.6	23.4	58.5	12.2	1.9	0	4
U2 - IPDE alpha = 0.9	25.8	57.7	11	0.8	0	4.7
U3 - IPDE alpha = 0	24.7	55.3	13	3.1	0.5	3.4
U3 - IPDE alpha = 0.3	25.2	57.8	10.3	2.3	0.2	4.2
U3 - IPDE alpha = 0.6	27.4	55.5	10.9	1.1	0.1	5
U3 - IPDE alpha = 0.9	30.5	56	7.6	0.7	0	5.2
Mean patients treated						
U2 - IPDE alpha = 0	8.4	11.7	5.7	3	0.9	
U2 - IPDE alpha = 0.3	8.9	12.2	5.4	2.6	0.7	
U2 - IPDE alpha = 0.6	9.5	12.5	5.1	2.2	0.4	
U2 - IPDE alpha = 0.9	10.3	12.6	4.8	1.8	0.3	
U3 - IPDE alpha = 0	9.1	11.4	5.4	3	0.8	
U3 - IPDE alpha = 0.3	9.5	12	5.2	2.4	0.6	
U3 - IPDE alpha = 0.6	10.1	12.3	4.9	2.1	0.4	
U3 - IPDE alpha = 0.9	10.9	12.4	4.5	1.7	0.2	
Mean IPDE patients						
U2 - IPDE alpha = 0	0.8	5	1.7	1.1	0.3	
U2 - IPDE alpha = 0.3	0.9	5.2	1.6	0.9	0.2	
U2 - IPDE alpha = 0.6	1	5.4	1.5	0.9	0.2	
U2 - IPDE alpha = 0.9	1	5.6	1.5	0.8	0.1	
U3 - IPDE alpha = 0	0.9	4.9	1.6	1.1	0.2	
U3 - IPDE alpha = 0.3	1	5.1	1.5	0.9	0.2	
U3 - IPDE alpha = 0.6	1	5.2	1.5	0.8	0.1	
U3 - IPDE alpha = 0.9	1.1	5.5	1.5	0.8	0.1	